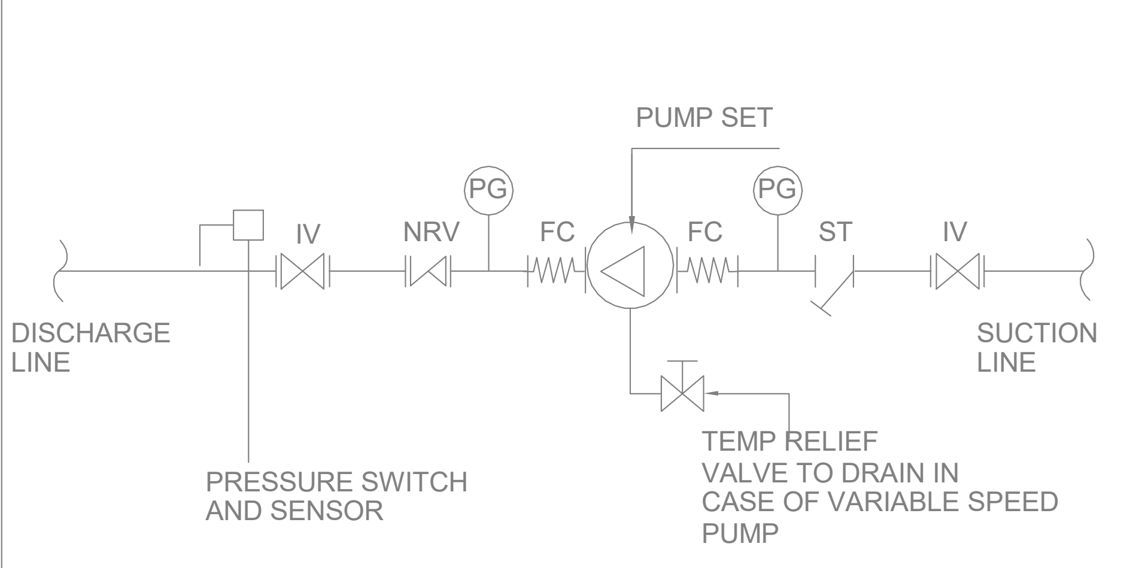
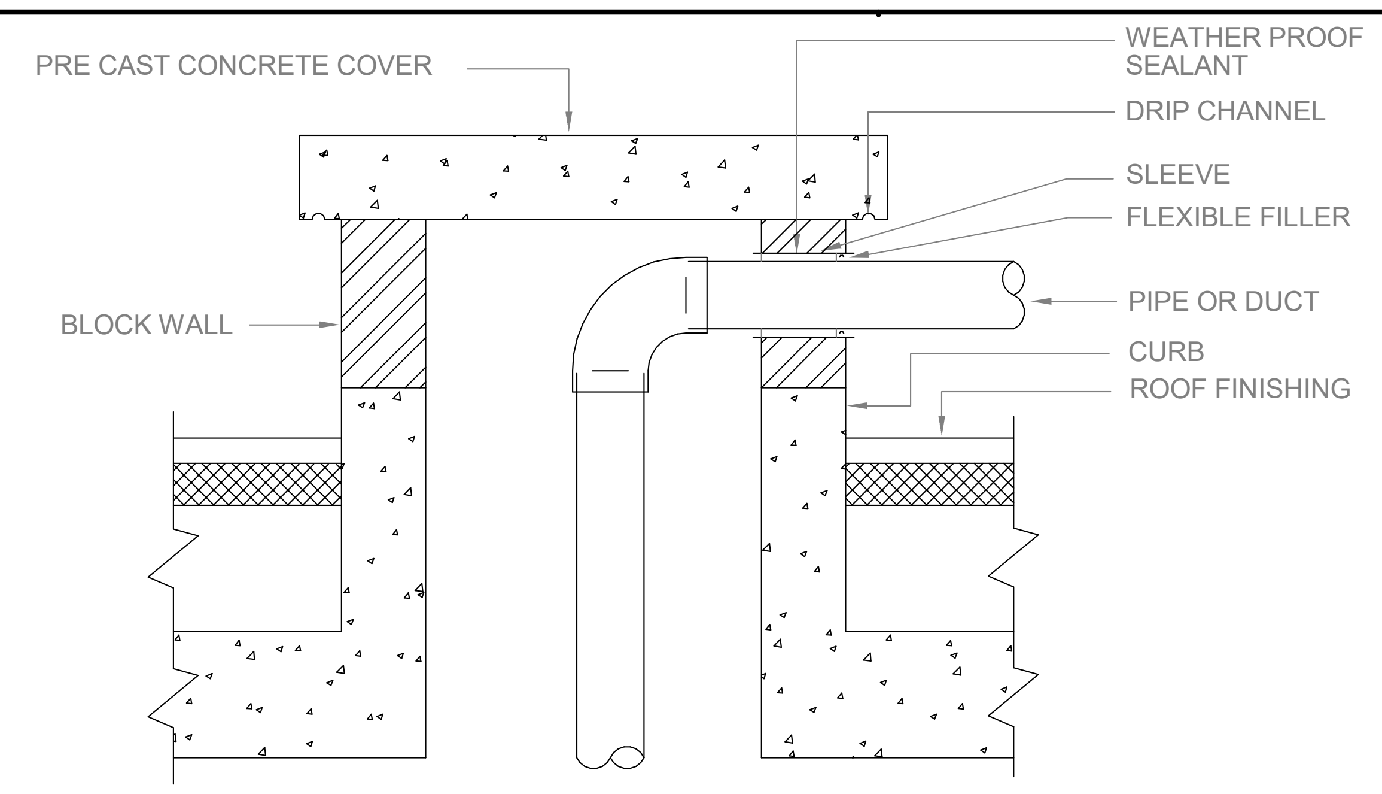
WATER SUPPLY SERVICES

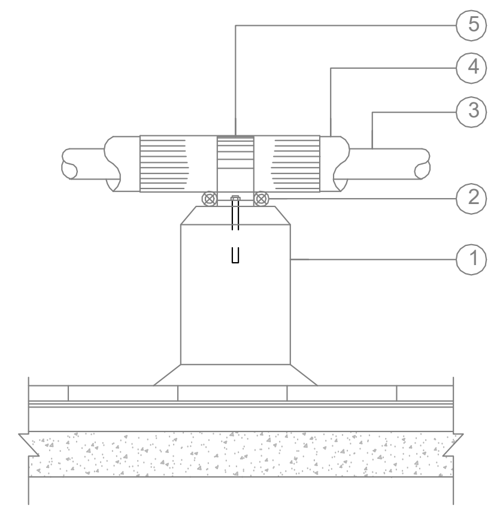

PUMP SET
PG
PG
IV
NRV
FC
FC
ST
IV
DISCHARGE LINE
SUCTION LINE
PRESSURE SWITCH AND SENSOR
TEMP RELIEF VALVE TO DRAIN IN CASE OF VARIABLE SPEED PUMP

NOTE:-
ALL PUMPS SHALL BE PROVIDED WITH DRY RUN PROTECTION

1 DETAIL TYPICAL BOOSTER PUMP CONNECTION DETAIL



2 DETAIL TYPICAL PIPE WORK ROOF PENTHOUSE

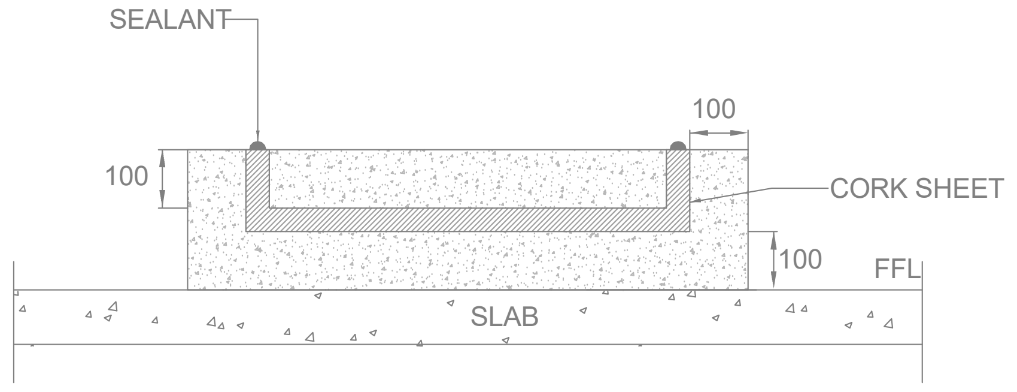


1. R.C / SOLID BLOCK SUPPORT WITH PLASTER & EPOXY
2. RUBBER PAD TO APPROVAL
3. ROOF WATER SUPPLY PIPE TO SPEC.
4. PIPE INSULATION .
5. FIXING CLAMP TO APPROVAL

3 DETAIL WATER SUPPLY PIPE ABOVE ROOF

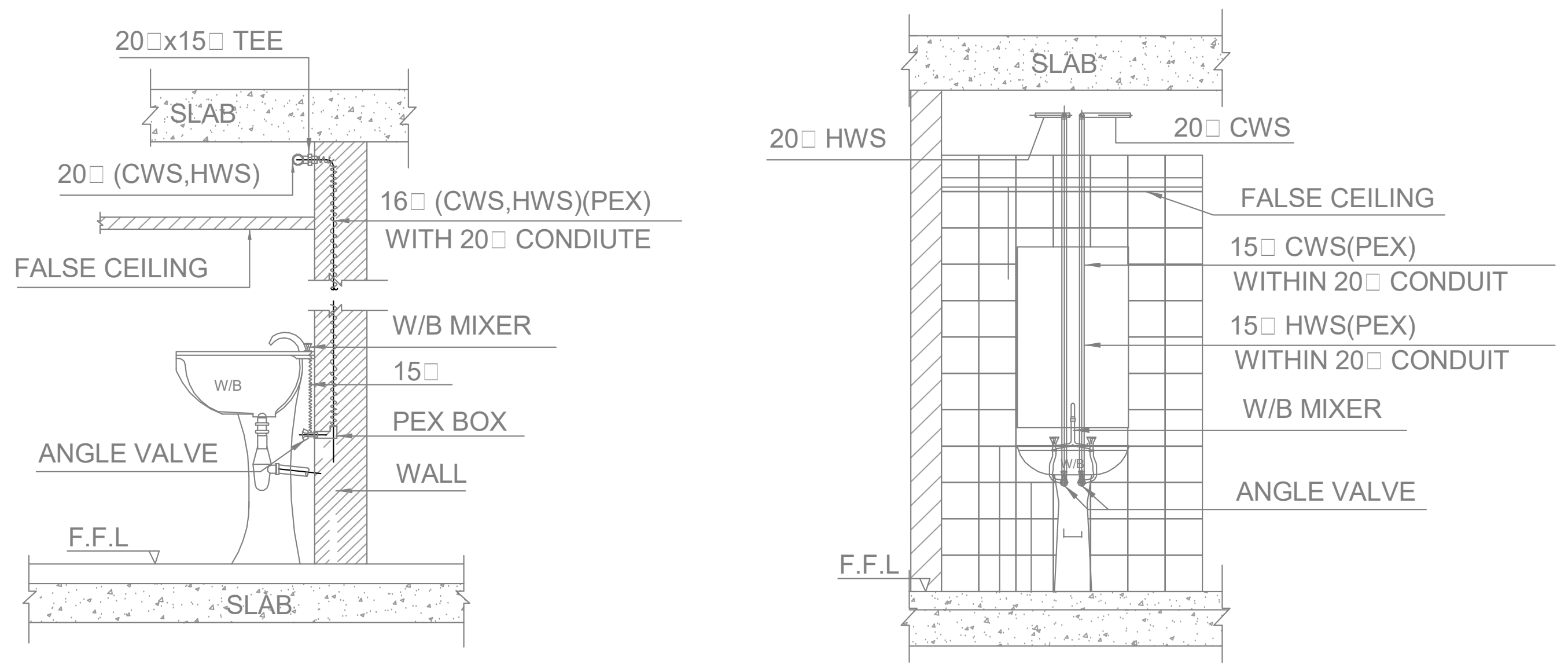
WATER NOTES

1. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE CO-ORDINATION OF ALL SERVICES.
2. THE PIPE SIZES ON DRAWINGS ARE BASED ON INTERNAL DIAMETER OF PIPE. CONTRACTOR TO PROVIDE OUTSIDE PIPES METER DIA ON SHOP DRAWINGS BASED ON SPECIAL PIPE MATERIAL.
3. UNLESS OTHERWISE INDICATED ON DRAWINGS, ALL PIPE WORK SHALL BE 20 MM DIA WITH REDUCTIONS TO 15MM□ FOR FINAL CONNECTION TO ALL FIXTURES SHALL BE 15 MM DIA.
4. FOR PIPE MATERIALS REFER MECHANICAL SPECIFICATION
5. EXPANSION COMPENSATOR SHALL BE PROVIDED ON ALL PIPE WORK AT CROSSING OF BUILDING EXPANSION JOINTS. PIPE SUPPORT SHALL BE PROVIDED BEFORE AND AFTER THE EXPANSION JOINT.
6. THE COLD AND HOT WATER VERTICAL PIPE DROP IN WALL CHASE SHALL BE IN PEX PIPE (CROSS-LINKED POLYETHYLENE) WITHIN CONDUIT.
7. ISOLATION VALVE SHALL BE PROVIDED ON THE MAIN COLD WATER BRANCH PIPE FOR EACH WET AREA(TOILET/KITCHEN/ETC.)
8. FINAL COLD AND HOT PIPE CONNECTIONS IN KITCHEN AND BATHROOM SHALL BE COORDINATED WITH THE INTERNAL EQUIPMENT LAYOUT.
9. ALL HIGH POINTS OF EVERY BOOSTED WATER PIPE RISER SHALL TERMINATE WITH AUTOMATIC AIR VENT. THE LOW POINT OF EACH RISER SHALL BE PROVIDED WITH DRAIN POINT WITH VALVE.
10. THE MECHANICAL EQUIPMENT/ PUMPS SHALL BE MOUNTED ON 150 MM CONCRETE PLINTH.
11. WATER TERMINATION HEIGHT FOR SANITARY WARES TO BE PROVIDED AS PER ARCHITECTURAL / ID DRAWINGS..

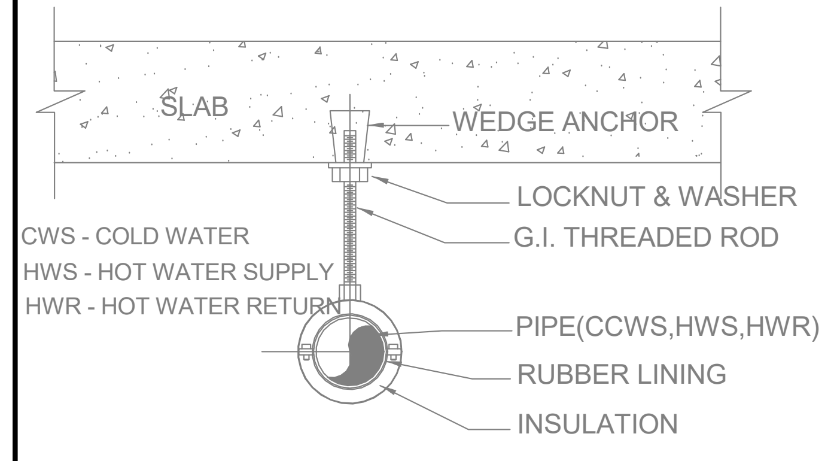


NOTE: ANTI VIBRATION MOUNTS SHALL BE PROVIDED FOR WATER PUMPS INSTALLATION ACCORDING TO ACOUSTIC DESIGN OF THE PROJECT.

4 DETAIL TYPICAL PUMP FOUNDATION DETAIL
(ABOVE HABITABLE SPACES)

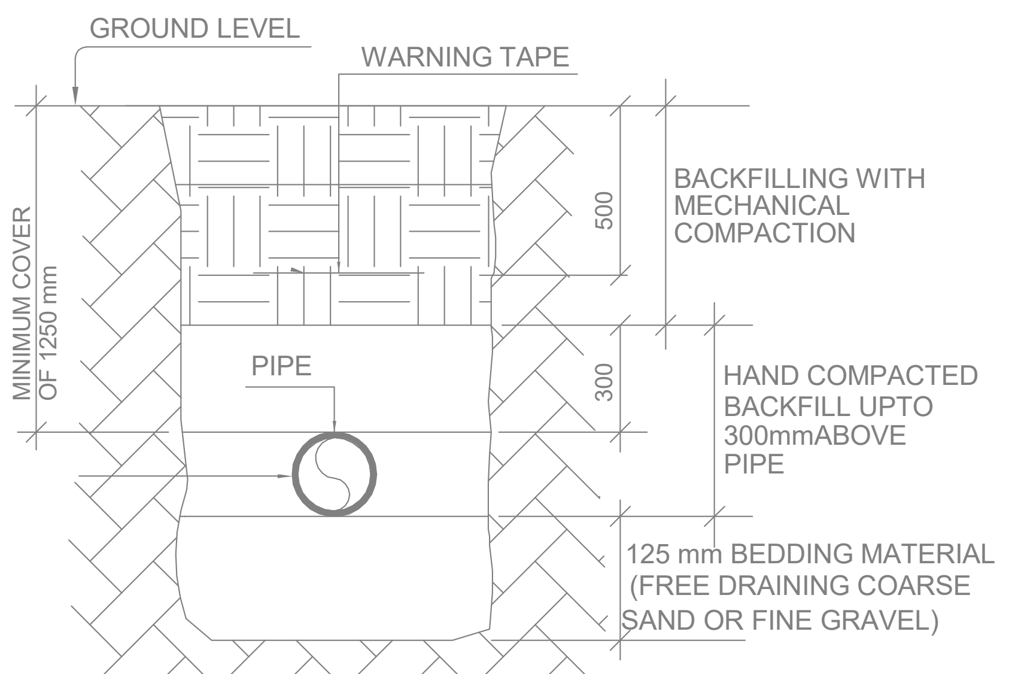


5 DETAIL WASH BASIN CONNECTION TYPICAL DETAIL



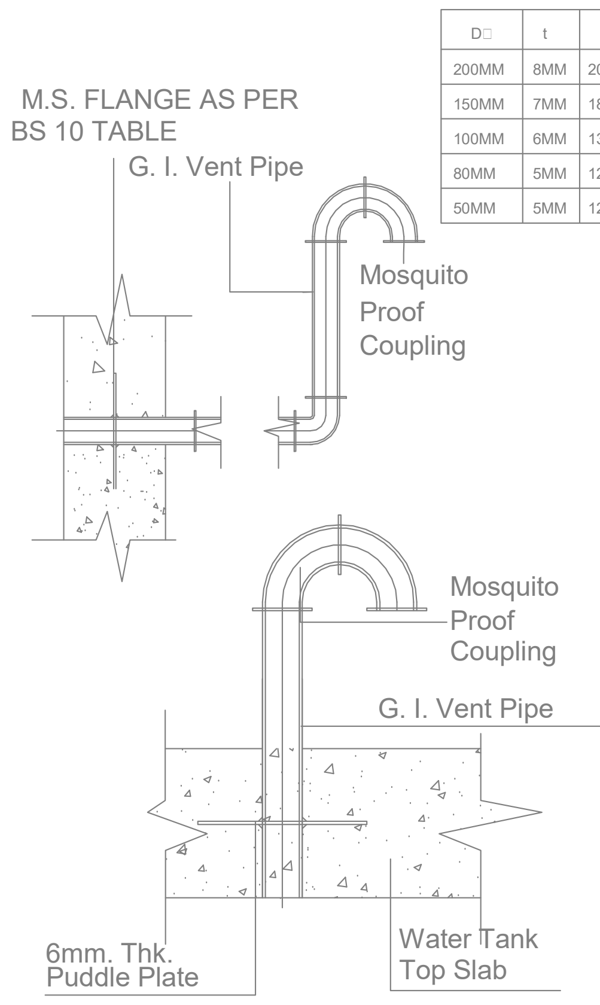
CWS - COLD WATER
HWS - HOT WATER SUPPLY
HWR - HOT WATER RETURN

6 DETAIL TYPICAL WATER SUPPLY PIPE(CCWS,HWS,HWR)
INSULATION & SUPPORTING DETAILS


GROUND LEVEL
WARNING TAPE
MINIMUM COVER OF 1250 mm
500
300
BACKFILLING WITH MECHANICAL COMPACTION
PIPE
HAND COMPACTED BACKFILL UPTO 300mmABOVE PIPE
125 mm BEDDING MATERIAL (FREE DRAINING COARSE SAND OR FINE GRAVEL)

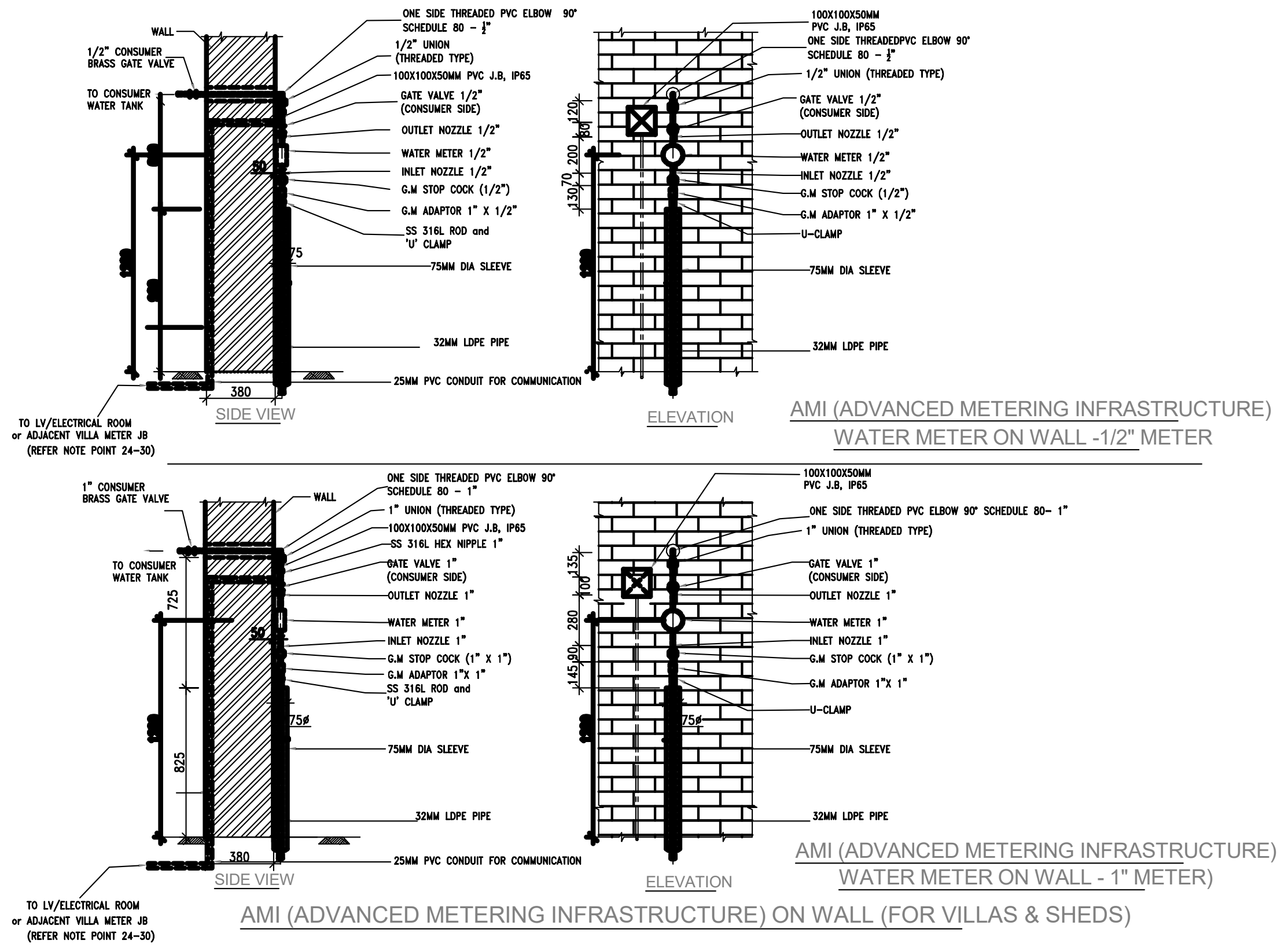
NOTE:-
THE CONTRACTOR TO FOLLOW PIPE MANUFACTURERS RECOMMENDATION FOR TRENCHING AND PIPE WORK INSTALLATION

7 DETAIL BACKFILL DETAIL FOR WATER MAINS



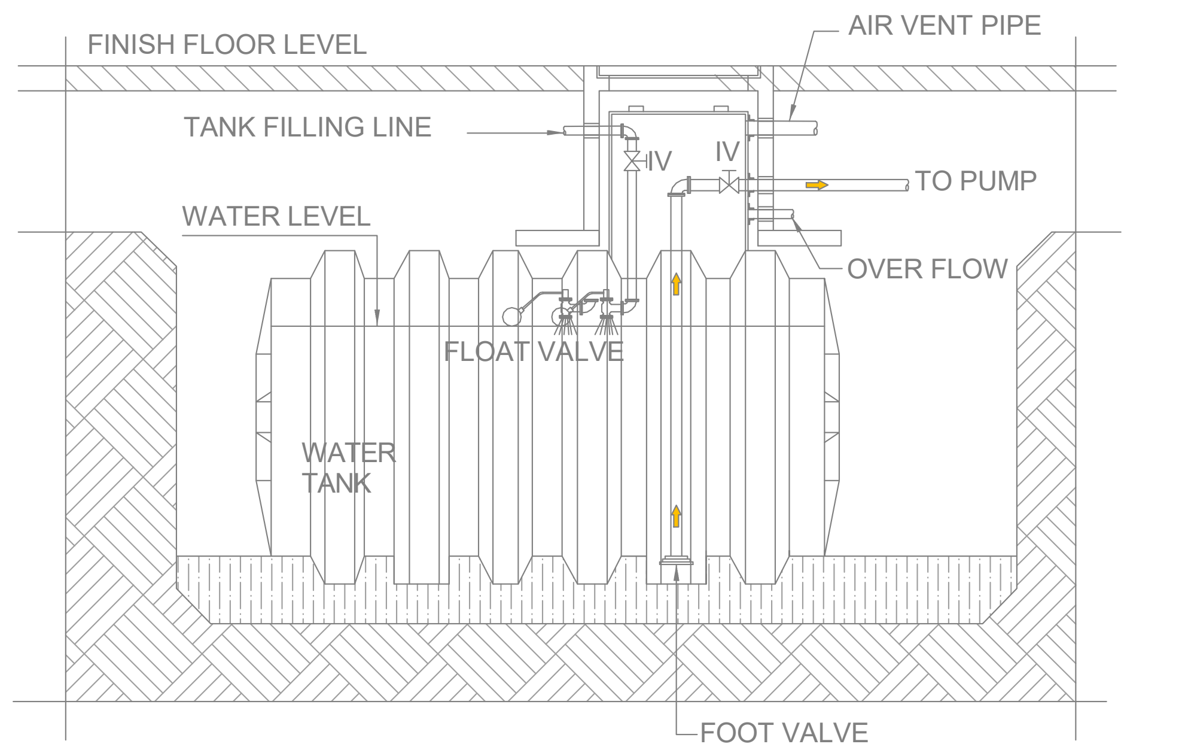
D□	t	T	d□
200MM	8MM	20MM	366MM
150MM	7MM	18MM	280MM
100MM	6MM	13MM	216MM
80MM	5MM	12MM	185MM
50MM	5MM	12MM	110MM

8 DETAIL DETAIL OF VENT PIPE FIXING (TYP.)

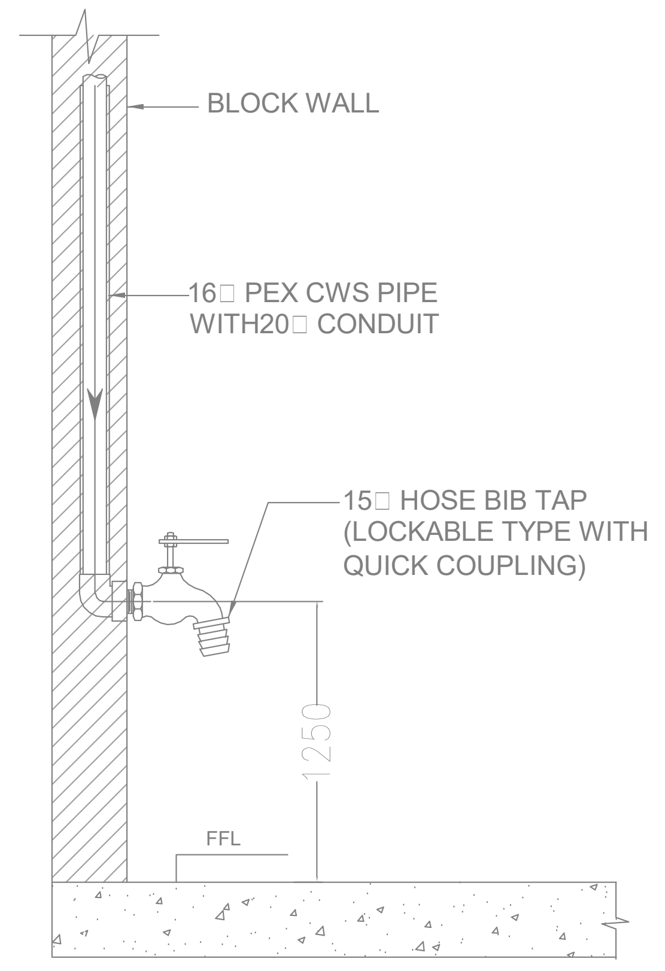

WALL
1/2" CONSUMER BRASS GATE VALVE
TO CONSUMER WATER TANK
ONE SIDE THREADED PVC ELBOW 90° SCHEDULE 80 - ½"
1/2" UNION (THREADED TYPE)
100X100X50MM PVC J.B, IP65
GATE VALVE 1/2" (CONSUMER SIDE)
OUTLET NOZZLE 1/2"
WATER METER 1/2"
INLET NOZZLE 1/2"
G.M STOP COCK (1/2")
G.M ADAPTOR 1" X 1/2"
SS 316L ROD and 'U' CLAMP
75MM DIA SLEEVE
32MM LDPE PIPE
25MM PVC CONDUIT FOR COMMUNICATION
50
75
380
TO LV/ELECTRICAL ROOM or ADJACENT VILLA METER JB (REFER NOTE POINT 24-30)
SIDE VIEW
100X100X50MM PVC J.B, IP65
ONE SIDE THREADEDPVC ELBOW 90° SCHEDULE 80 - ½"
1/2" UNION (THREADED TYPE)
GATE VALVE 1/2" (CONSUMER SIDE)
OUTLET NOZZLE 1/2"
WATER METER 1/2"
INLET NOZZLE 1/2"
G.M STOP COCK (1/2")
G.M ADAPTOR 1" X 1/2"
U-CLAMP
75MM DIA SLEEVE
32MM LDPE PIPE
120
80
200
70
130
ELEVATION
AMI (ADVANCED METERING INFRASTRUCTURE) WATER METER ON WALL -1/2" METER
1" CONSUMER BRASS GATE VALVE
WALL
TO CONSUMER WATER TANK
ONE SIDE THREADED PVC ELBOW 90° SCHEDULE 80 - 1"
1" UNION (THREADED TYPE)
100X100X50MM PVC J.B, IP65
SS 316L HEX NIPPLE 1"
GATE VALVE 1" (CONSUMER SIDE)
OUTLET NOZZLE 1"
WATER METER 1"
INLET NOZZLE 1"
G.M STOP COCK (1" X 1")
G.M ADAPTOR 1"X 1"
SS 316L ROD and 'U' CLAMP
75ø
75MM DIA SLEEVE
32MM LDPE PIPE
25MM PVC CONDUIT FOR COMMUNICATION
725
825
50
380
TO LV/ELECTRICAL ROOM or ADJACENT VILLA METER JB (REFER NOTE POINT 24-30)
SIDE VIEW
100X100X50MM PVC J.B, IP65
ONE SIDE THREADED PVC ELBOW 90° SCHEDULE 80- 1"
1" UNION (THREADED TYPE)
GATE VALVE 1" (CONSUMER SIDE)
OUTLET NOZZLE 1"
WATER METER 1"
INLET NOZZLE 1"
G.M STOP COCK (1" X 1")
G.M ADAPTOR 1"X 1"
U-CLAMP
75ø
75MM DIA SLEEVE
32MM LDPE PIPE
135
100
280
90
145
ELEVATION
AMI (ADVANCED METERING INFRASTRUCTURE) WATER METER ON WALL - 1" METER)

AMI (ADVANCED METERING INFRASTRUCTURE) ON WALL (FOR VILLAS & SHEDS)

9 DETAIL WATER METER DETAILS



10 DETAIL POLYETHYLENE WATER TANK DETAIL
(CONSTRUCTION AS PER MANUFACTURER'S RECOMMENDATION)


BLOCK WALL
16□ PEX CWS PIPE WITH20□ CONDUIT
15□ HOSE BIB TAP (LOCKABLE TYPE WITH QUICK COUPLING)
1250
FFL

TYPICAL DETAILS OF BIB TAP CONNECTION
(CONCEALED TYPE)

11 DETAIL TYPICAL DETAILS OF BIB TAP CONNECTION
(CONCEALED TYPE)

EQUIVALENT PIPE DIAMETERS

NOMINAL INSIDE PIPE DIA(mm)	UPVC PIPE DIA(mm)	COPPER PIPE DIA(mm)	POLYPROPYLENE (PPR-PN-16) (mm)	POLYPROPYLENE (PPR-PN-20) (mm)
15	15	15	20	25
20	20	22	25	32
25	25	28	32	40
32	32	35	40	50
40	40	42	50	63
50	50	54	63	75
65	65	67	75	90
80	80	76	90	110
100	100	108	125	160
150	150	159	200	200

EQUIPMENT PIPE DIAMETERS

S.NO	Fixture Type	Requirements	Remarks
01	Water Closets	3/6 l/flush	Dual flush type
02	Urinals	1.0 l/flush or waterless	Sensor based urinal systems are also acceptable.
03	Wash Basin Mixture/ Faucets/Sink Mixer/ Bath Mixer	6.0 l/min.	At 413.7 Kpa reference pressure
04	Shower Heads	8.0 l/min.	At 551.6 Kpa reference pressure
05	Kitchen Sink	7.0 l/min.	At 413.7 Kpa reference pressure
06	Dishwasher	4 l/min.	

MINIMUM PERFORMANCE CRITERIA FOR SANITARY WARES

Key Plan

N

General Notes :

01. Do not Scale The Drawings. Follow (The MM) Written Dimensions Unless Otherwise Mentioned.
02. Each Drawing Shall be Read in Conjunction With all Other Drawings,Specification & the General Standard Details. In Case of Any Discrepancies Bring it to the Knowledge of Engineer and Same to be Resolved Prior to Execution of Work.
03. Dimensions Shown on These Drawings are Structural Dimensions. Unless Noted Otherwise.
04. All Levels Indicated On These Drawings are Levels as per the the Survey Datum, Same Should be Verified at Site Prior to Execution of Work.
05. All Levels Indicated On These Drawings are Based On Design Level, Which is Equivalent to Datum Level and Same Should be Verified at Site Prior to Execution of Work.

TENDER DRAWINGS

No.	Date	Initials	Description
00	10/03/2026	LSG	TENDER SUBMISSION
00	16/01/2026	LSG	DETAILED DESIGN SUBMISSION
00	31/10/2025	LSG	SCHEMATIC DESIGN SUBMISSION
00	15/10/2025	LSG	SCHEMATIC SUBMISSION-50%
00	01/09/2025	LSG	CONCEPT DESIGN SUBMISSION

REVISIONS / ISSUE

THIS DRAWING MUST NOT BE REPRODUCED IN WHOLE OR IN PART WITHOUT THE PRIOR APPROVAL OF ARIF & BINTOAK

Status
TENDER SUBMISSION

Client
EMAAR

Lead Consultant
الأستشاري:
عارف وبن طوق
Arif&Bintoak
consulting architects & engineers
POST BOX 3670 DUBAI.UAE TEL 04-3372888 FAX 04-3370352
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Project Title:
Proposed Residential Townhouses
and Standalone Villas
TH-1, TH-2, and TH-3 at Grand Polo
Club & Resort, Dubai, UAE

Discipline : MECHANICAL

Drawing title:
WATER SUPPLY STANDARD DETAILS

Architectural Engineer	Electrical Engineer	Structural Engineer
MUSTAFA ABID	MARWAN MOHD.	RIHAM RAMADAN
Registration # 5847	Registration # 80102	Registration # 105092
Scale(s): NTS	Date : 10/03/2026	Project no: 1857
Drawn by: LSG	Checked : MAV	Approved : MRM

Drawing Number : GPCR-TH02-AB-DRG-ZZZ-MEC-WS-4001-00

16-Feb-26 3:15:06 PM

This 2D drawing is extracted from 3D BIM Model

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